

Felt Productivity Is Not Measured Productivity

Why AI adoption should be judged on delivered output, not on how fast it feels

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EXECUTIVE SUMMARY

Before scaling an AI tool, compare its delivered output with a baseline taken before the tool arrived, and keep self-reported speed as a separate record.

Most people will name a confident figure for how much AI speeds them up, but the people using AI are poor witnesses to their own speed. A program scaled on that number is scaled on sentiment. A small measurement standard separates the two: a baseline captured before the tool arrives, an output measure that heavier tool use cannot inflate, and a review that reads felt speed and delivered output side by side.

- 01** Treat self-reported speedup as sentiment, not evidence: in a randomized trial, developers believed AI sped them up 20 percent while the measured work ran 19 percent slower.
- 02** Leave adoption charts, prompt counts, and survey speedup out of the evidence; they show that people use and like the tool, not that the output improved.
- 03** When the team feels faster but delivered output has not moved, that gap is the finding the review exists to catch, not an anomaly to explain away.
- 04** Capture the baseline and fund the measurement now: as AI use spreads, the before-AI comparison the standard depends on gets harder to obtain.

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They felt faster. The clock said otherwise.

Ask people how much faster AI made them. The answer is almost always a confident number. Here is the finding that should unsettle it. In a 2025 randomized controlled trial, the research nonprofit METR had sixteen experienced open-source developers complete 246 real tasks. They worked in their own

repositories, with and without AI help. Going in, the developers expected the tools to speed them up by 24 percent. Coming out, having lived through the work, they still believed AI had sped them up by 20 percent. The clock said the opposite. They took 19 percent longer with the tools than without [1].

EXHIBIT 1

Capable people misread their own speed

**19% vs
20%**

In METR's randomized trial, experienced developers were 19 percent slower with AI, yet afterward still believed it had sped them up by 20 percent — the gap this standard governs.

METR randomized controlled trial: 16 experienced open-source developers, 246 real tasks, July 2025.

What the trial shows is narrow, and the limits matter. It does not show that AI makes people slower. It measured sixteen expert developers, on mature codebases they knew well, using early-2025 tooling, at one moment in time. Change any of those conditions and the number can move. Read it as a verdict on the tools and you repeat the exact error it exposes.

What the trial does show is durable. Capable people believed one thing about their own speed, and the clock recorded another. The belief survived direct contact with the evidence.

The same split shows up across a whole delivery system. Google's DORA program found the same pattern in its 2024 Accelerate State of DevOps report. AI adoption "significantly increases individual productivity, flow, and job satisfaction" while it "negatively impacts software delivery stability and throughput." [3] Read those two findings together. People felt more productive and were more satisfied. The delivery system they worked in became less stable and pushed less work through. Individual experience and system output pointed in opposite directions, in the same dataset.

Neither study is the last word. Both are snapshots, and METR's authors have since said as much. But together they name the thing a governance standard has to account for, and it is not the direction of any one result. How fast the work feels and what the work delivers are two separate things, and the first is not a reliable stand-in for the second.

People are poor witnesses to their own speed

A feeling cannot govern because the witnesses are unreliable and do not know it. METR's own follow-up, published in February 2026, is blunt about the self-report problem. The developers' estimates of their own speedup, it notes, "can be quite unreliable." [2] That is not a claim that people lie. The tool answers instantly. The draft appears. The blank page fills. That sensation is simply a poor instrument for measuring elapsed time and reworked output. It feels like progress even when review, correction, and re-prompting follow and erase the saving.

Set the forecast beside the result and the mechanism shows. A 24 percent expected gain is optimism about a new capability. The 20 percent remembered gain is the more troubling figure. It was reported after the work was done, when experience was supposed to correct the estimate. It did not. If living through the work does not correct the number, a survey asking employees how much AI helped is measuring something else. It is measuring enthusiasm, tool preference, and the natural wish to have spent one's effort well. Not help.

There is a second reason self-report cannot carry a governance decision, and it is getting worse rather than better. METR's follow-up reports that the "wider adoption of AI has made it more difficult to measure task-level productivity" [2]. The comparison against unaided work is eroding. In that follow-up, 30 to 50 percent of developers chose not to submit some tasks, because they did not want to do them without AI [2]. The honest reading is uncomfortable for anyone who wanted a clean number. The more normal AI use becomes, the harder it is to isolate its effect. And the harder that gets, the more tempting one number becomes. It is always available and always flattering: what people say about themselves.

A third problem is built into the survey format itself. A survey samples memory, and memory is exactly what failed in the trial. A group that had just been slowed down, on average, remembered being sped up. Worse, the people an organization asks about a tool are more likely to be the ones who chose it, championed it, or enjoy it. The sample is not neutral, and the question invites the answer.

So the direction of the evidence matters more than any single figure. The durable claim is not about speed at all. Self-reported speedup is a low-quality signal that systematically runs ahead of measured output, and an organization scaling AI on that signal is scaling on sentiment. A portfolio does not fund sentiment on purpose. It does so by default, when no one built the alternative.

What to measure instead: three things

The alternative is ordinary. It is the same discipline any organization uses to measure a change in how the work gets done. AI simply arrived wrapped in enough novelty to make people forget they already knew how. NIST's AI Risk Management Framework makes measurement one of its four core functions. MEASURE uses "quantitative, qualitative, or mixed-method" tools to analyze and monitor AI risk and

related impacts [5]. Where trade-offs among trustworthy characteristics arise, "measurement provides a traceable basis" to inform management decisions [5]. Measurement works here as a control, not as an after-the-fact report. It lets a leader decide whether to rely on the system more, hold, or stop.

So the standard this paper proposes is narrow and testable. Three things have to exist before an AI-assisted workflow is being governed on evidence. A baseline, captured before the tool changed the work. A work-output signal that reflects delivered value rather than activity. And a review that reads the felt experience and the delivered outcome as two separate records. A program can show a rising adoption chart and a glowing internal survey and still not produce those three. It is then being run on impression, however disciplined the rollout looked.

Take the before-picture first

A CIO feature on AI's ROI caught the most common failure in one line from a former global CIO: "We didn't start with a baseline." [6] Without a before-picture, no after-picture can be checked. Any improvement can be credited to the tool. Any decline can be explained away. There is no honest record of how the work performed when a person did it unaided. The baseline does not need to be elaborate. It needs to be captured before the workflow changes, on the same measure the review will use afterward. A baseline rebuilt from memory after adoption inherits exactly the optimism this whole problem is made of.

Count finished work, not activity

A baseline is only as good as what it measures, and this is where most AI dashboards go wrong. They count activity: seats provisioned, prompts sent, tokens consumed, features accepted. Activity is easy to instrument and always goes up. The SPACE framework was built to correct that reflex. Researchers at GitHub, Microsoft, and the University of Victoria published it in ACM Queue. Developer productivity, they argue, is about "more than an individual's activity levels" and "cannot be measured by a single metric or dimension." [4] SPACE names five dimensions: satisfaction and well-being, performance, activity, communication and collaboration, and efficiency and flow. For a workflow outside software, the useful move is to ask which of the five an AI metric actually touches. Adoption charts and prompt counts are activity. Survey speedup is satisfaction. Only performance and, sometimes, efficiency describe whether the work came out better. Those two are the ones instrumentation usually skips, because they are the hardest to capture.

The lesson holds outside software too. A work-output signal answers a question about delivered value: cycle time on a real case, rework rate, error rate, coverage, or throughput of finished units that met standard. It is chosen so that using the tool more cannot move it by itself. Usage is an input. Output is what the portfolio funded.

Keep the feeling and the result in separate columns

The evidence demands this third element most directly. The review keeps two columns and never collapses them. One holds what people report about the experience of the work. The other holds what the work actually did against the baseline. Both are legitimate. Satisfaction and flow are real, they matter for keeping people and for adoption, and DORA found AI genuinely raised them [3]. But they are recorded as experience, next to the output signal, not merged into it. When the two columns agree, reliance can grow with confidence. When they diverge — the team feels faster while the delivered work has not moved, or has slipped — that divergence is the most valuable thing the review produces. It is precisely the case a satisfaction survey alone would have scored as a win.

Which numbers count as evidence, and which do not

The hard part is deciding which numbers do not count. Adoption rate, active-user counts, prompts per employee, self-rated speedup: all easy to collect, and all pointing the wrong way. They measure that AI is being used and that people enjoy using it, not that the work improved. They belong on an adoption dashboard, where the question is whether a rollout reached people. They do not belong in the evidence a portfolio uses to decide whether to deepen reliance. None of them can tell a workflow that got better from one that merely got busier.

The signals worth keeping share one property. They live where the value is actually created or lost. A contributed CIO column on the AI ROI gap put this exactly. The governance machinery of "steering committees, approval gates and KPI dashboards" sits above the work, while "the value, or the leak, happens inside the workflow, at the level of the actual task the AI is doing." You can, the same piece warns, "instrument your governance structure perfectly and still have nothing measuring whether the agent's output was right at the point where it mattered." [7] That is the difference between a measure that governs and one that reassures. It is also why a gain on one task may not reach the bottom line. In separate ROI reporting, a law-firm CIO noted there is "no straight line between the investment in that tool and increased revenues" [6].

Two cautions bound the measurement itself. First, measurement often goes missing because no one sets it up. A KPMG AI leader in the same reporting said companies need to track AI use and "how it correlates to work output," adding, "that measurement is fundamentally lacking" [6]. Measuring is work that has to be funded. It does not appear on its own once a tool is bought. Second, the instinct to measure harder can curdle into surveillance, and surveillance damages the very work it watches. The goal is a small set of honest output signals tied to a baseline, owned by someone accountable for the workflow. Not a monitoring apparatus that counts keystrokes and erodes the trust that adoption depends on.

What it costs to scale on the wrong number

This can look like a measurement nicety. It is not. The cost of governing on felt speed is real, and it lands on the portfolio. The first cost is misdirected capacity. When a scaling decision rests on a survey, the tools and workflows that produce the strongest feelings attract the next round of investment. Whether they produced the strongest results does not enter into it. Scarce integration, change, and review capacity follows the enthusiasm rather than the evidence. The organization compounds its bets on the wrong thing precisely because the wrong thing felt best.

The second cost is credibility, and it works slowly. A program promoted on a promised speedup eventually meets the results it was supposed to move. When the delivered numbers do not show the gain the survey implied, the gap does not read as a measurement error. It reads as a broken promise, and it spends the sponsor's credibility and the wider appetite for AI investment along with it.

The third cost arrives later. Savings claimed on felt speed are sometimes cashed in advance — in a headcount plan, a benefits case, or a budget cut justified by an efficiency no one measured. A saving that existed only as a feeling still has to be paid back, in rework, escalation, and quiet backfill, once the work fails to deliver it. Measuring the work is cheaper than any of these. It is cheapest of all before the commitment, not after the shortfall.

How the review works when the columns disagree, and when they agree

Here is the shape the standard takes when a real review runs it. This is an operating pattern, not any one organization. A team adopts an AI assistant across a recurring, high-volume workflow — the kind where a small saving per case would compound into a real number. Adoption is fast and the sentiment is excellent. The internal survey reports a large felt speedup, and the case for expanding the license writes itself. Under an impression-led process, that is where the decision gets made.

Under this standard, the survey is one column, and it is not the column that decides. The review pulls the work-output signal the team agreed on before rollout: cycle time on completed cases that met the quality bar. It compares that against the baseline captured from the pre-tool period. In the pattern I have seen hold up, the two columns disagree more often than sponsors expect. The felt speedup is genuine. The measured cycle time on completed, standard-meeting cases has barely moved. The measurement does not say by itself where the saving went, only that it did not reach the delivered outcome. That is the finding. A real, local time saving stopped somewhere short of the result the portfolio funded, and only the output signal made the shortfall visible. Where exactly it stopped is then the workflow owner's question to run down.

That disagreement is not a reason to abandon the tool. It is the review doing its job. It changes the next decision from "expand it, because people love it" to "the value is stuck in the workflow, not the tool —

redesign the review step, then measure again." The felt speed told the team something true about the experience of the work. The output signal told them something true about the delivery of it. Only holding both let them act on the difference instead of funding past it.

The standard does more than slow things down. Sometimes the two columns agree: the team reports a genuine speedup, and the output signal on completed, standard-meeting work has moved with it against the baseline. Then the review produces the opposite verdict, and produces it with evidence a skeptical finance partner can inspect. That is the case for deepening reliance, expanding the license, and moving the workflow from trial to standing operation. Separating the columns does not mean distrusting every good feeling. It makes the yes as well evidenced as the no. Expansion then rests on delivered work, not on how persuasive the people who liked the tool happened to be.

The one-page standard: seven things to name

The discipline should fit on a single page and survive a change of sponsor. Then, when a scaling decision arrives, the evidence is already defined and only the judgment remains. A workable standard names seven things before an AI-assisted workflow is scaled.

EXHIBIT 2

A one-page standard names seven things before a workflow is scaled

- 1 Named workflow and owner**
The recurring workflow under review and the person accountable for its delivered outcome, not for the tool.
- 2 Pre-tool baseline**
The output measure captured before the tool changed the work, on the same definition the review will reuse.
- 3 Work-output signal**
A delivered-value measure — cycle time, rework, error rate, finished units — that using the tool more cannot move by itself.
- 4 Felt-experience record**
Satisfaction, flow, and self-reported speed, captured honestly and kept in their own column, never merged into output.
- 5 The separation review**
A standing check that reads the two columns side by side and treats a divergence between them as the primary finding.
- 6 The reliance decision**
An explicit outcome each cycle — deepen, hold, redesign the workflow, or stop — tied to the signal and baseline, not the adoption chart.
- 7 The measurement cost**
A funded commitment to instrument the workflow, because an unfunded standard quietly reverts to self-report.

A standard of this shape does not slow adoption. It protects it. It lets a leader say yes to deeper reliance on evidence, and lets them redirect rather than abandon a tool whose value is leaking inside the workflow. Most of all it prevents the specific failure the research warns about: rolling a capability out across an organization on the strength of how fast it felt.

What these studies do and do not show

Honesty about the evidence is part of the standard. A measurement discipline that overstated its own proof would refute itself. The METR trial is rigorous but bounded in exactly the ways its own section set out, and it licenses no general claim that AI reduces productivity. This paper makes none. DORA's finding carries a different limit. It is an association across a survey population, not a controlled experiment, and delivery stability and throughput are themselves reported by the respondents. It is also a 2024 snapshot of a fast-moving practice [3]. Confidence comes from their agreement on the

shape of the problem, not from either number on its own. Felt experience and measured output diverge, and the divergence favors the feeling.

EXHIBIT 3

What the evidence establishes, and what it does not

-  **METR trial**
 A randomized trial shows a large gap between felt speed and measured output that survived the trial.

-  **DORA 2024**
 System-level snapshot: adoption raised satisfaction as delivery stability and throughput fell.

-  **SPACE framework**
 Productivity is multidimensional; no single metric or activity count captures it.

-  **NIST AI RMF**
 Measurement is an operating control that gives management decisions a traceable basis.

-  **Not established**
 No universal metric, review schedule, or benefit multiplier for AI; both anchors are snapshots.

The deeper caution comes from METR's own follow-up. Wider adoption is making task-level measurement harder [2]. The clean baseline this standard depends on is getting more expensive to capture, not less. An organization that waits may find the unaided comparison has vanished from its own operations. That is an argument for building the measurement discipline now, while a before-picture is still possible, rather than a reason to defer it until the tools "settle." No source here establishes a universal work-output metric, a required review schedule, or a benefit multiplier for AI. Those are properties of a specific workflow, and the standard is deliberately a method for finding them rather than a set of numbers to adopt.

The test that matters

The test of AI governance is not the enthusiasm of a rollout or the confidence of a survey. Any capable tool will produce both, and produce them early. The real test comes when the felt speed and the delivered outcome disagree: can the organization see the disagreement, and act on the outcome? An

enterprise that measures the work — a baseline, a real output signal, and a review that keeps the feeling and the result in separate columns — is governing its AI. An enterprise that scales on how fast the work feels is governing its mood, and funding it as if it were evidence.

Felt productivity is real, and it is worth having. It is simply not the same thing as measured productivity, and only one of the two can tell a portfolio where its money went.

Notes and sources

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About the author

Marco Policani is an enterprise portfolio, PMO, and AI operating-governance leader. He builds portfolio governance systems where AI carries the analytical load and named humans own every decision — an approach documented across case studies, walkthroughs, and working governance guides on his portfolio site.

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